

# One Skill Does Not Fit All: Automatic Discovery and Taxonomy-Guided Routing of Frame-Selection Skills for Long-Video Question Answering

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## Abstract

Long-Video Question Answering (LVQA) requires locating decisive visual evidence from videos spanning tens of minutes or even hours under a limited frame budget. Since relevant evidence may appear only briefly or across distant segments, effective frame selection is crucial. Existing training-free methods typically rely on a single frame-selection mechanism for all questions. However, our empirical analysis shows that different selection skills specialize in different semantic question categories, and their relative effectiveness also varies across target benchmarks. This suggests that LVQA should move beyond searching for one universal selector toward automatically discovering a diverse skill toolbox and adapting its use to the target domain. We present **AutoSkill**, a training-free framework that automatically discovers complementary frame-selection skills and performs target-adaptive, taxonomy-guided skill routing. Starting from a small labelled source pool, LLM agents iteratively propose, implement, evaluate, and refine executable selection skills. AutoSkill induces a shared semantic taxonomy from pooled target-domain adaptation queries. It then resamples semantically compatible labelled source examples and rewrites their questions and options into the target style while preserving the original video grounding and answer labels. Evaluating the discovered skills on this target-style routing set yields a category-to-skill mapping. At inference time, AutoSkill assigns each test question to a semantic category and executes only its associated frame-selection skill, retaining approximately the inference cost of a single-skill method. Across five long-video benchmark splits, AutoSkill improves Qwen2.5-VL-7B and Qwen3.5-4B by 2.4% and 1.0% respectively, without updating model parameters or using target answer annotations.

## Introduction

Long-Video Question Answering (LVQA) requires answering natural language questions by locating relevant visual evidence from videos spanning tens of minutes or even hours. Such videos arise in many real-world scenarios, including lectures, movies, and egocentric recordings (Zhou et al. 2025; Wu et al. 2024; Fu et al. 2025; Wang et al. 2025b). The evidence required to answer a question may appear briefly,

recur across distant segments, or depend on the overall evolution of an event. Exhaustively processing all frames is impractical, as the resulting visual tokens quickly exceed the context capacity of current Multimodal Large Language Models (MLLMs) (Zhang et al. 2024a; Chen et al. 2025b; Shu et al. 2025). Therefore, effective LVQA requires selecting a small set of informative frames under a limited visual token budget.

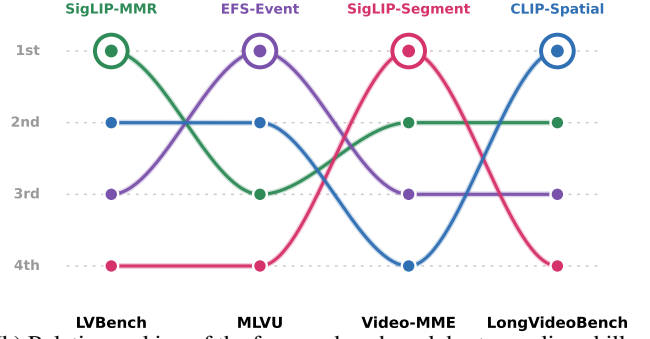
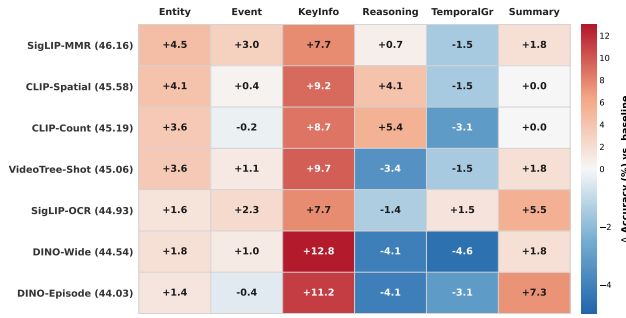
Existing approaches address this challenge from two directions: improving MLLMs through additional training or reducing visual inputs before inference. Training-based methods (Zhang et al. 2024a; Chen et al. 2025b; Shen et al. 2024; Shu et al. 2025) extend context windows, compress visual tokens, or learn temporal representations. Although effective, they require substantial training resources and are often tied to specific model architectures. Training-free approaches (Tang et al. 2025; Wang et al. 2025c; Sun et al. 2025; Zhang et al. 2025b; Guo et al. 2026; Zou et al. 2025b) provide a more flexible alternative by selecting informative frames through mechanisms such as query-guided retrieval, temporal localization, coarse-to-fine search, and iterative evidence refinement. These methods can be directly applied to frozen MLLMs with low overhead. However, existing approaches typically rely on a single manually designed frame-selection strategy, implicitly assuming that one selection rule can generalize across diverse LVQA questions and target domains.

We argue that this assumption is overly restrictive. Different questions require different forms of visual evidence acquisition, and consequently different frame-selection mechanisms may be preferable. For example, temporal reasoning questions may benefit from localized search, while counting or entity-related questions may require broader coverage or specialized sampling patterns. As shown in Fig. 1, our empirical analysis reveals complementary specialization of frame-selection skills at two levels. Within a benchmark, different question categories favour different skills. Across benchmarks, the relative ranking of skills also changes, with no single skill consistently dominating all target distributions. These observations suggest that instead of designing one universal selector, LVQA requires automatically discovering a diverse toolbox of selection skills and adapting their usage according to the target question distribution.

To this end, we propose **AutoSkill**, a training-free framework that automatically discovers complementary frame-

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(a) Per question type accuracy gains ( $\Delta$ ) over uniform sampling on LVBench. (b) Relative ranking of the four per-benchmark best sampling skills.

**Figure 1: No single frame-selection skill consistently dominates.** Within a benchmark, the best-performing skill depends on the question type, while across benchmarks, the relative ranking of skills changes substantially, with each of four benchmarks selecting a different top performer. This diversity indicates that complementary evidence is dispersed across skills at both granularities, motivating taxonomy-guided category-to-skill routing rather than committing to a fixed selection strategy.

selection skills and performs target-adaptive taxonomy-guided skill routing. AutoSkill consists of three stages (Fig. 2). In Stage 1, AutoSkill uses an LLM agent to automatically discover a compact toolbox of frame-selection skills. Starting from a shared labelled source pool, the agent iteratively proposes, implements, executes, and evaluates candidate selection programs. Execution feedback guides subsequent iterations toward more effective and diverse skills, without requiring manual design of the complete skill library. In Stage 2, AutoSkill adapts the discovered toolbox to a target LVQA domain without accessing target evaluation answers. It first induces a semantic taxonomy from target-domain questions, capturing different types of evidence required by target queries. Based on this taxonomy, AutoSkill constructs a target-style routing set by selecting semantically compatible labelled source examples and rewriting their questions and options to match the target query style while preserving the original video grounding and answer labels. Evaluating different skills on this adapted routing set produces a category-to-skill performance matrix, from which a skill is assigned to each semantic category. In Stage 3, AutoSkill applies the learned category-to-skill mapping during inference. Each test question is first assigned to a semantic category, and only the corresponding frame-selection skill is executed before the frozen MLLM generates the final answer. Unlike exhaustive skill execution or parameterized router training, AutoSkill adapts the selection mechanism while retaining approximately the inference cost of a single-skill method.

**Our main contributions are as follows:**

1. We reveal that a single frame-selection strategy is insufficient for long-video QA. Through systematic analysis, we show that different selection skills exhibit complementary strengths across question categories and target benchmarks, motivating adaptive skill composition.

2. We propose **AutoSkill**, a training-free framework that automatically discovers a diverse frame-selection skill toolbox through LLM-agent-driven iterative exploration and execution feedback. AutoSkill further adapts the discovered toolbox to target LVQA domains through taxonomy-guided category-to-skill assignment without accessing target evalu-

ation answers.

3. We introduce an efficient category-to-skill routing mechanism that selects one appropriate skill for each semantic question category and avoids executing the entire toolbox at inference. Across four long-video QA benchmarks covering five evaluation splits, AutoSkill improves frozen Qwen2.5-VL-7B and Qwen3.5-4B by 2.4% and 1.0% on average, respectively, without model updates or target answer annotations.

## Related Work

**Efficient Long-Video Understanding.** Long-video understanding is constrained by temporal redundancy and limited visual tokens. Training-based methods address this through long-context adaptation, token compression, hierarchical representations, and external memory (Zhang et al. 2024a; Chen et al. 2025b; Shen et al. 2024; Shu et al. 2025), but they usually need video-language supervision and stay tied to one backbone. Training-free methods instead reduce the visual input before inference via query relevance (Zhu et al. 2025; Park et al. 2026; Zhang et al. 2025b), visual diversity (Tang et al. 2025; Peng et al. 2026), event localization (Ye et al. 2025; Zou et al. 2025b; Guo et al. 2026), or iterative search (Ye et al. 2025; Guo et al. 2026; Park et al. 2026). These approaches are strong and inexpensive, yet most optimize one selection mechanism and apply it to all questions. Query conditioning can change frame scores, but the evidence-acquisition strategy itself remains fixed. AutoSkill automatically discovers complementary frame-selection skills and learns when each skill should be applied.

**Automatic Skill Discovery and Adaptive Routing.** Adaptive computation appears in mixture of experts (Shazeer et al. 2017; Fedus, Zoph, and Shazeer 2022), model cascades (Chen, Zaharia, and Zou 2023; Ong et al. 2024), and LLM tool use (Schick et al. 2023; Qin et al. 2024; Patil et al. 2024). Most systems assume an existing expert or tool library, and their routers often need supervised trajectories, learned retrievers, reward optimization, or multi-expert execution (Ong et al. 2024; Zheng et al. 2026; Zou et al. 2025a;

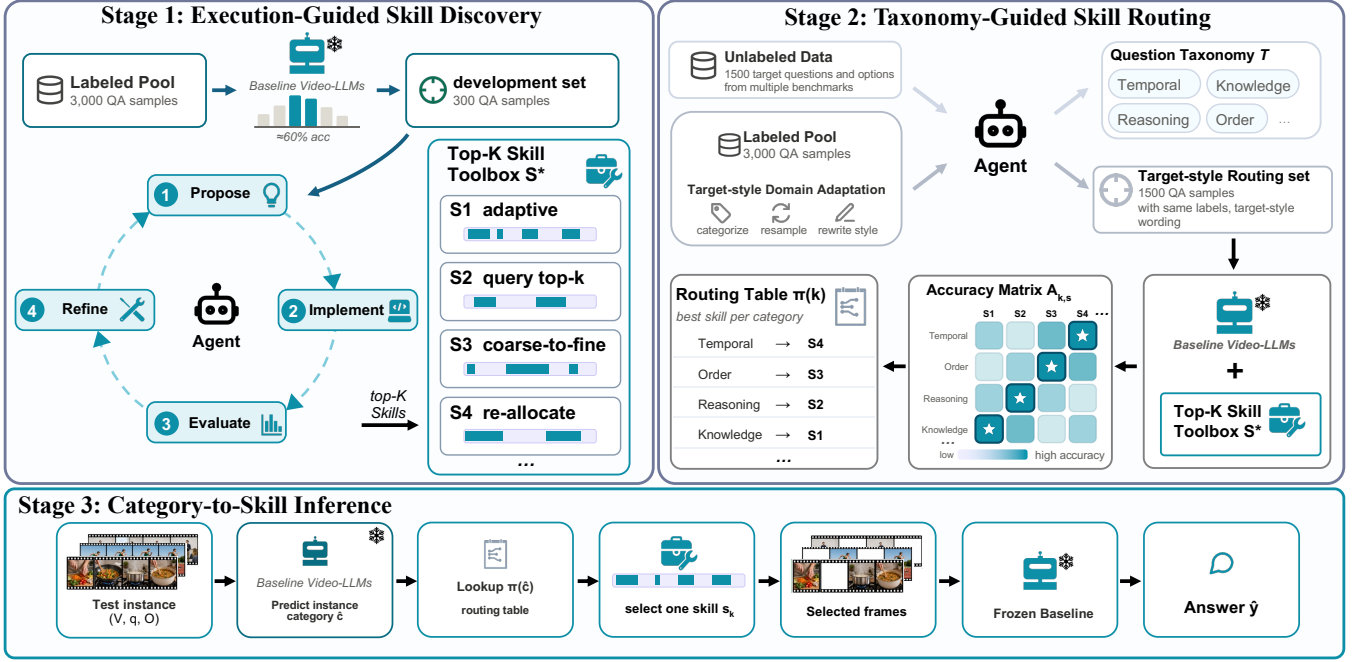


Figure 2: Overview of AutoSkill. Stage 1 iteratively discovers a Top- $K$  skill toolbox on a capability-focused development set. Stage 2 constructs a category-to-skill routing table from unlabelled target questions and a target-style adaptation of the shared labelled pool. Stage 3 predicts the question category at test time, looks up one skill, and answers with a single frozen MLLM pass, without using test labels.

Chen et al. 2025a). Agent systems can request and refine tools during solving (Fei, Zheng, and Feng 2025; Zou et al. 2025a), but rarely build a reusable video skill toolbox from downstream evaluation. AutoSkill brings automatic skill discovery into long-video frame selection: an LLM agent discovers frame-selection skills on a capability-focused development set and distills their execution history into a category-to-skill mapping without parameter updates or target benchmark answers.

## Methodology

### Problem Formulation

We study long-video question answering (LVQA), where the goal is to answer a question  $q$  given a long video  $V$ . Due to the limited context capacity of frozen MLLMs, only a subset of frames can be provided as visual input. Let  $B$  denote the frame budget. A frame-selection skill  $s$  determines how to allocate this budget by selecting input frames:

$$\mathcal{F}_s = s(V, q), \quad |\mathcal{F}_s| \leq B, \quad (1)$$

The frozen MLLM predicts the answer based on the selected frames:

$$\hat{y}_s = f_\theta(\mathcal{F}_s, q, \mathcal{O}), \quad (2)$$

Existing training-free approaches typically apply one predefined frame-selection mechanism to all questions. In contrast, AutoSkill aims to automatically discover a toolbox  $\mathcal{S}^*$  of complementary selection skills and assign suitable skills to different semantic question categories.

Specifically, given a target-domain question taxonomy  $\mathcal{T} = \{\tau_1, \dots, \tau_K\}$ , AutoSkill learns a category-to-skill mapping:

$$\pi : \mathcal{T} \rightarrow \mathcal{S}^*, \quad (3)$$

where each category is associated with a suitable frame-selection skill. During inference, each test question is assigned to a semantic category  $\tau \in \mathcal{T}$ , and only the corresponding skill  $\pi(\tau)$  is executed.

### Capability-Focused Development Set Construction

Automatic skill discovery requires an informative evaluation environment that can reveal the strengths and limitations of different frame-selection mechanisms. However, randomly sampled LVQA data often contains many samples that are either trivially solved or consistently failed by frozen MLLMs, providing limited signals for distinguishing the effectiveness of different selection strategies.

To provide informative feedback for skill discovery, we construct a capability-focused development set from a shared labelled pool. Specifically, we first collect  $P = 3,000$  labelled samples from VideoVista (Li et al. 2024b), AL-LVB (Tan et al. 2025), and LongVideo-Reason (Chen et al. 2026). The pool is disjoint from all evaluation benchmarks, and no model parameters are updated using these samples. This shared pool is used only for automatic skill discovery and target-style routing construction.

From this pool, we select a subset of 300 intermediate-difficulty samples such that the frozen base MLLM achieves

approximately 60% accuracy:

$$\mathcal{D}_{\text{dev}} = \{(V_i, q_i, y_i)\}_{i=1}^{N_{\text{dev}}}, \quad N_{\text{dev}} = 300. \quad (4)$$

Compared with randomly sampled data, these samples provide a balanced mixture of solvable and challenging cases, allowing the discovery agent to observe both successful and failed selection behaviors. Such informative feedback helps identify where improved frame-selection strategies can benefit the frozen MLLM.

To preserve diverse temporal and semantic characteristics, the development set is constructed through uniform stratification over video-duration groups and task categories. We further examine different development set sizes and observe that increasing the set size from 300 to 450 or 1,000 samples provides limited additional improvement (Tab. 4). Therefore, we use 300 samples as a compact and computationally efficient environment for iterative skill exploration.

### Execution-Guided Skill Discovery

A key challenge in constructing a skill-based frame-selection system is that the effective and complementary selection mechanisms are unknown in advance. Different LVQA questions may require different forms of evidence acquisition, including temporal localization, long-range coverage, or visual prioritization. Therefore, instead of manually designing a fixed toolbox of selection strategies, AutoSkill uses an LLM agent to automatically discover a diverse set of executable frame-selection skills.

Specifically, we use the capability-focused development set  $\mathcal{D}_{\text{dev}} = \{(V_i, q_i, y_i)\}_{i=1}^{N_{\text{dev}}}$  as a closed-loop evaluation environment for skill discovery. The development set contains intermediate-difficulty samples where the frozen MLLM exhibits non-trivial failures while still providing meaningful execution feedback, enabling the discovery agent to identify the strengths and limitations of different frame-selection mechanisms.

The discovery process follows an iterative execution-guided exploration paradigm. In each cycle, the agent analyzes previous skill executions and error patterns, proposes new frame-selection mechanisms or refinements, implements executable skills, and evaluates them on the development set. The resulting performance feedback is then used to guide subsequent exploration. Unlike searching for a single globally optimal selector, this process aims to construct a compact and diverse toolbox in which different skills capture different evidence acquisition patterns, without requiring a manually predefined skill library.

Each discovered skill  $s$  defines a mechanism for allocating the fixed frame budget. Given a video-question pair  $(V, q)$ , the skill selects the input frames:

$$\mathcal{F}_s = s(V, q), \quad |\mathcal{F}_s| \leq B, \quad (5)$$

where  $B$  denotes the maximum number of frames provided to the frozen MLLM. The selected frames are then processed by the same frozen model:

$$\hat{y}_s = f_\theta(\mathcal{F}_s, q, \mathcal{O}), \quad (6)$$

All candidate skills share the same backbone model, question prompt, frame budget, and inference configuration. Therefore, the performance differences mainly reflect the effectiveness of different frame-selection mechanisms.

For each skill, we evaluate its quality using the development-set accuracy:

$$R(s; \mathcal{D}_{\text{dev}}) = \frac{1}{N_{\text{dev}}} \sum_{i=1}^{N_{\text{dev}}} \mathbb{I}[f_\theta(s(V_i, q_i), q_i, \mathcal{O}_i) = y_i], \quad (7)$$

Beyond overall accuracy, we maintain fine-grained statistics across video duration, question category, and data source. These statistics provide the agent with detailed execution feedback and reveal the complementary strengths of different selection mechanisms. After multiple exploration cycles, we retain a compact toolbox of effective and diverse skills, denoted as  $\mathcal{S}^*$ , for subsequent target-domain adaptation.

### Target-Adaptive Taxonomy-Guided Skill Routing

The discovered toolbox  $\mathcal{S}^*$  contains complementary frame-selection skills, but their effectiveness varies across target domains with different question distributions and evidence requirements. Applying all skills to every query introduces substantial inference overhead, while selecting a single fixed skill ignores the specialization of different skills across question categories. Therefore, AutoSkill constructs a target-adaptive category-to-skill mapping using target-domain query semantics and a target-style routing set, without accessing target evaluation samples or answers.

**Query Taxonomy Induction.** We first characterize the semantic structure of target-domain questions without using target answers. Given unlabelled target-domain questions from multiple benchmarks, we sample question-option pairs:

$$\mathcal{Q}_{\text{tgt}} = \bigcup_{j=1}^J \{(q_{j,i}, \mathcal{O}_{j,i})\}_{i=1}^M. \quad (8)$$

An LLM agent analyzes these query-option pairs and induces a semantic taxonomy  $\mathcal{T} = \{\tau_1, \dots, \tau_{K_q}\}$ . Each category represents a type of evidence requirement, such as temporal localization, event ordering, counting, or global understanding. The taxonomy captures question semantics rather than benchmark identity. Each target query is assigned to one semantic category:

$$c_i = A_{\text{cat}}(q_i, \mathcal{O}_i; \mathcal{T}), \quad c_i \in \{1, \dots, K_q\}, \quad (9)$$

where  $A_{\text{cat}}$  denotes taxonomy-based semantic categorization. This step only uses unlabelled target-domain questions and options, and no target evaluation answers are accessed.

**Target-Style Routing Set Construction.** The induced taxonomy defines the organization of target questions but does not directly reveal which skill is suitable for each category. To establish this correspondence, we reuse the shared labelled pool introduced in Section . For each target category, we retrieve source examples with compatible evidence requirements according to the induced taxonomy. The selected examples are then rewritten to match the linguistic style and

Table 1: Comparison on long video understanding benchmarks. <sup>†</sup> marks Qwen2.5-VL-7B post-training methods whose scores are taken from ParaVT (Yang et al. 2026b). <sup>‡</sup> denotes our reimplementaion under the 128-frame budget.

| Model                                                                    | Size | MLVU        |             | LongVideoBench<br>val | VideoMME    | LVBench     | Avg/ $\Delta$    |
|--------------------------------------------------------------------------|------|-------------|-------------|-----------------------|-------------|-------------|------------------|
|                                                                          |      | dev         | test        |                       |             |             |                  |
| Proprietary Models                                                       |      |             |             |                       |             |             |                  |
| GPT-4o (Hurst et al. 2024)                                               | -    | 64.6        | 54.9        | 66.7                  | 71.9        | 48.9        | —                |
| Gemini-2.5-Pro (Comanici et al. 2025)                                    | -    | 81.2        | -           | -                     | 84.3        | 78.7        | —                |
| Open-Source VLMs                                                         |      |             |             |                       |             |             |                  |
| Video-LLaVA (Lin et al. 2024a)                                           | 7B   | 36.2        | 30.7        | 37.6                  | 39.9        | -           | —                |
| LLaVA-OneVision (Li et al. 2024a)                                        | 7B   | 64.7        | 47.2        | 56.4                  | 58.3        | -           | —                |
| LLaVA-NeXT-Video (Zhang et al. 2024b)                                    | 7B   | -           | -           | 49.1                  | -           | -           | —                |
| Video-XL (Shu et al. 2025)                                               | 7B   | 64.9        | 45.5        | 50.7                  | 55.5        | -           | —                |
| VILA-1.5 (Lin et al. 2024b)                                              | 7B   | 56.7        | -           | -                     | -           | -           | —                |
| Kangaroo (Liu et al. 2026)                                               | 8B   | 61.0        | -           | 54.8                  | 56.0        | 39.4        | —                |
| LongVU (Shen et al. 2024)                                                | 7B   | 65.4        | -           | -                     | 60.6        | -           | —                |
| LongVA (Zhang et al. 2024a)                                              | 7B   | 56.3        | 41.1        | -                     | 52.6        | -           | —                |
| LongVILA (Chen et al. 2025b)                                             | 7B   | -           | -           | 57.1                  | 60.1        | -           | —                |
| Qwen2.5-VL (Bai et al. 2025)                                             | 7B   | 64.4        | 50.7        | 58.7                  | 64.0        | 42.3        | —                |
| LongVILA-R1 (Chen et al. 2026)                                           | 7B   | -           | -           | 57.6                  | 62.4        | -           | —                |
| Video-R1 (Feng et al. 2026)                                              | 7B   | 65.0        | 49.2        | 52.0                  | 61.1        | 38.7        | —                |
| VideoZoomer (Ding et al. 2025)                                           | 7B   | 68.8        | 55.8        | 57.7                  | 65.2        | 41.5        | —                |
| VideoChat-R1 <sup>†</sup> (Li et al. 2025)                               | 7B   | 58.7        | -           | 49.2                  | 50.4        | 23.8        | —                |
| ReWatch-R1 <sup>†</sup> (Zhang et al. 2025a)                             | 7B   | 60.1        | -           | 53.6                  | 58.8        | 38.5        | —                |
| Video-Thinker <sup>†</sup> (Wang et al. 2025a)                           | 7B   | 65.2        | -           | 56.0                  | 61.9        | -           | —                |
| Conan <sup>†</sup> (Ouyang et al. 2026)                                  | 7B   | 59.2        | -           | 54.5                  | 55.5        | 38.2        | —                |
| LongVT-RFT <sup>†</sup> (Yang et al. 2026a)                              | 7B   | 59.4        | -           | 54.7                  | 59.5        | 37.9        | —                |
| SAGE <sup>†</sup> (Jain et al. 2026)                                     | 7B   | 49.7        | -           | 37.4                  | 44.1        | 31.8        | —                |
| Discovered Skill Toolbox (Ours, only selected Top-5 skills) <sup>‡</sup> |      |             |             |                       |             |             |                  |
| Qwen2.5-VL <sup>‡</sup> (Bai et al. 2025)                                | 7B   | 66.0        | 47.7        | 61.2                  | 64.6        | 42.5        | 56.4/—           |
| SigLIP-MMR                                                               | 7B   | 68.3        | 51.5        | 61.6                  | <b>65.3</b> | 46.2        | 58.6/+2.2        |
| CLIP-Spatial                                                             | 7B   | 68.8        | <b>52.1</b> | <b>62.0</b>           | 64.0        | 45.6        | 58.5/+2.1        |
| CLIP-Count                                                               | 7B   | 67.7        | 50.6        | <b>62.0</b>           | 64.1        | 45.2        | 57.9/+1.5        |
| CLIP-MMR                                                                 | 7B   | 67.6        | 51.2        | 61.1                  | 64.5        | 44.9        | 57.9/+1.5        |
| SigLIP-OCR                                                               | 7B   | 67.4        | 50.1        | 61.5                  | <b>65.3</b> | 44.9        | 57.8/+1.4        |
| AutoSkill (ours) <sup>‡</sup>                                            | 7B   | <b>69.0</b> | 51.1        | 61.9                  | 65.2        | <b>46.8</b> | <b>58.8/+2.4</b> |

option format of the target domain while preserving the original video grounding, evidence requirements, and answer labels. This produces a target-style routing set:

$$\mathcal{D}_{\text{route}} = \{(V_i, \tilde{q}_i, \tilde{\mathcal{O}}_i, y_i, c_i)\}_{i=1}^N. \quad (10)$$

Here,  $\tilde{q}_i$  and  $\tilde{\mathcal{O}}_i$  denote the rewritten query and options, while  $y_i$  remains unchanged. Therefore, the routing set provides target-style evaluation signals for skill comparison without requiring target evaluation annotations.

**Category-to-Skill Assignment.** We evaluate every discovered skill  $s \in \mathcal{S}^*$  on the target-style routing set. For each semantic category, we compute the accuracy of every skill and select the best-performing one:

$$\pi(k) = \arg \max_{s \in \mathcal{S}^*} A_{k,s}, \quad (11)$$

where  $A_{k,s}$  denotes the accuracy of skill  $s$  on routing samples assigned to category  $k$ . The resulting mapping  $\pi : \mathcal{T} \rightarrow \mathcal{S}^*$  associates each semantic question category with a suitable frame-selection skill.

## Inference with Category-to-Skill Routing

At inference time, AutoSkill first assigns each question-option pair to a semantic category according to the induced taxonomy:

$$\hat{c} = A_{\text{cat}}(q, \mathcal{O}; \mathcal{T}), \quad (12)$$

where  $\hat{c}$  denotes the predicted semantic category. The corresponding frame-selection skill is retrieved from the category-to-skill mapping:

$$\hat{s} = \pi(\hat{c}). \quad (13)$$

The selected skill then determines the input frames under the fixed budget:

$$\hat{\mathcal{F}} = \hat{s}(V, q), \quad |\hat{\mathcal{F}}| \leq B. \quad (14)$$

The frozen MLLM produces the final answer based on the selected frames:

$$\hat{y} = f_{\theta}(\hat{\mathcal{F}}, q, \mathcal{O}). \quad (15)$$

Only one frame-selection skill is executed for each query, preserving approximately the inference cost of a single-skill

Table 2: Qwen3.5-4B baseline, Top-5 discovered skills, and AutoSkill on long-video understanding benchmarks. <sup>†</sup> denotes our reimplementaion under the 128-frame budget.

| Model / Skill         | Size | MLVU |      | LongVideoBench<br>val | VideoMME | LVBench | Avg/ $\Delta$ |
|-----------------------|------|------|------|-----------------------|----------|---------|---------------|
|                       |      | dev  | test |                       |          |         |               |
| Qwen3.5 <sup>†</sup>  | 4B   | 69.9 | 55.1 | 61.3                  | 67.3     | 42.3    | 59.2/–        |
| QMSD-Zoom             | 4B   | 69.5 | 55.8 | 62.2                  | 65.8     | 44.6    | 59.6/+0.4     |
| Boundary-Evidence     | 4B   | 69.2 | 55.4 | 62.1                  | 64.9     | 45.1    | 59.3/+0.1     |
| Indexed-Evidence      | 4B   | 68.9 | 55.5 | 62.3                  | 66.1     | 46.3    | 59.8/+0.6     |
| Clip-Face-Expression  | 4B   | 69.9 | 55.5 | 62.3                  | 66.0     | 46.1    | 60.0/+0.8     |
| Count-Diversity-Nudge | 4B   | 69.9 | 55.4 | 62.4                  | 67.0     | 42.3    | 59.4/+0.2     |
| AutoSkill (ours)      | 4B   | 69.4 | 57.5 | 62.4                  | 67.1     | 45.4    | 60.4/+1.2     |

method while adapting the selection mechanism to different semantic question categories.

## Experiments

### Experimental Setup

**Benchmarks.** We evaluate on four long-video QA benchmarks covering five evaluation splits: MLVU (dev and test) (Zhou et al. 2025), LongVideoBench (val) (Wu et al. 2024), Video-MME (w/o subtitles) (Fu et al. 2025), and LVBench (Wang et al. 2025b), using each benchmark’s official evaluation protocol and a unified frame budget of  $B=128$ .

**Implementation details.** Routing uses  $J=5$  benchmarks with  $M=300$  unlabelled question–option pairs each ( $|\mathcal{Q}_{\text{tgt}}|=1,500$ ),  $K_q=19$  induced categories, and a rewritten routing set of  $N=1,500$  samples from the  $P=3,000$  labelled pool in Section . Backbones are frozen Qwen2.5-VL-7B (Bai et al. 2025) and Qwen3.5-4B (Qwen Team 2026) with  $B=128$  and a direct MCQ prompt. Discovery is run by a Claude Code agent with Auto Research (Yang, Li, and Li 2026), with a Codex reviewer who critiques proposals before coding. Skills may call auxiliaries of at most 1B parameters (e.g., CLIP, SigLIP, DINOv2, Grounding DINO) and may change only frame allocation. Following Table 3, we run  $T=6$  cycles with  $M_c=5$  evaluations each. Cycle 1 proposes 5 skills, and later cycles use 3 refinements plus 2 new explorations. Failed reviews are dropped before GPU runs. Runtime errors fall back to uniform sampling. Skills that regress versus uniform are excluded from  $\mathcal{S}^*$ . No human filters or edits accepted skills.

### Main Results

**Results with Qwen2.5-VL-7B.** Table 1 compares AutoSkill with the frozen backbone, individual discovered skills, and existing proprietary and open-source models. Under our unified 128-frame budget, the Qwen2.5-VL-7B baseline obtains an average accuracy of 56.4% across the 5 benchmark splits. Every Top-5 discovered skill improves this average, with absolute accuracy gains ranging from 1.4% to 2.2%, confirming that the agent repeatedly discovers useful strategies rather than a single isolated improvement. Their strengths are nevertheless complementary: CLIP-Spatial performs best on both

Table 3: Skill discovery across successive feedback cycles on the development set.  $\Delta$  is relative to uniform sampling. We use  $T=6$  in the main experiments after observing saturation.

| Variant                  | Avg. acc. (%) | $\Delta$ |
|--------------------------|---------------|----------|
| Baseline                 | 60.0          | –        |
| Best skill after Cycle 1 | 61.0          | +1.0     |
| Best skill after Cycle 4 | 62.7          | +2.7     |
| Best skill after Cycle 6 | 63.0          | +3.0     |
| Best skill after Cycle 8 | 63.0          | +3.0     |

MLVU splits and ties for the best LongVideoBench score among individual skills, whereas SigLIP-MMR is strongest on LVBench and ties for the best VideoMME score. By routing instances among these specialized strategies, AutoSkill reaches 69.0%, 51.1%, 61.9%, 65.2%, and 46.8% on MLVU-dev, MLVU-test, LongVideoBench, VideoMME, and LVBench, respectively. This yields the best average among our Qwen2.5-VL variants, improving the frozen baseline by 2.4%. AutoSkill improves over the baseline on every split and obtains the strongest overall average, even when it does not top every individual benchmark, demonstrating that taxonomy-guided routing converts skill complementarity into consistent overall gains.

**Results with Qwen3.5-4B.** Table 2 evaluates discovery with frozen Qwen3.5-4B instead of transferring the Qwen2.5-VL toolbox, and obtains separate Top-5 skills. The Qwen3.5-4B baseline averages 59.2%. Individual discovered skills improve this by 0.1–0.8%, and AutoSkill reaches 60.4% (+1.2%), showing that the same discovery-and-routing protocol adapts to a different model capability profile.

### Ablation Studies

**Execution feedback accumulates useful refinements.** As shown in Table 3, the best skill after Cycle 1 improves Qwen2.5-VL-7B by 1.0%. Continued diagnosis and refinement raise the gain to 2.7% after Cycle 4 and 3.0% after Cycle 6. Extending the search to Cycle 8 yields no further improvement on the development set. Execution feedback therefore continues to help through early cycles and then saturates, so we set  $T=6$  for the main experiments.

**Development-set size.** Table 4 compares discovery on

Table 4: Development-set size ablation on Qwen2.5-VL-7B. Accuracies are on the same capability-focused set.  $\Delta$  is vs. uniform sampling.

| $N_{\text{dev}}$ | Baseline (%) | Best skill (%) | $\Delta$ |
|------------------|--------------|----------------|----------|
| 300              | 60.0         | 63.0           | +3.0     |
| 450              | 54.4         | 58.4           | +4.0     |
| 1,000            | 49.9         | 52.9           | +3.0     |

Table 5: Router comparison averaged on four long-video QA benchmarks covering five evaluation splits. Majority voting aggregates the Top-5 skills. DLR and AutoSkill select one skill and require one video MLLM pass.

| Strategy                       | Avg. acc. (%) | $\Delta$    | Passes |
|--------------------------------|---------------|-------------|--------|
| Baseline                       | 56.4          | –           | 1      |
| Best single skill              | 58.6          | +2.2        | 1      |
| Discovery Cycle Router (DLR)   | 57.4          | +1.0        | 1      |
| Majority voting                | 58.6          | +2.2        | 5      |
| <b>AutoSkill router (ours)</b> | <b>58.8</b>   | <b>+2.4</b> | 1      |

capability-focused sets of 300, 450, and 1,000 samples. Absolute baseline accuracy decreases as the set grows harder, but the best discovered skill improves the baseline by +3.0%, +4.0%, and +3.0%, respectively. Enlarging the development set therefore does not yield a substantially larger discovery gain, while evaluation cost scales roughly with  $N_{\text{dev}}$ . We use  $N_{\text{dev}}=300$  in the main experiments.

**Router comparison.** As shown in Table 5, DLR improves uniform sampling by 1.0%. The final taxonomy-guided router reaches 58.8%, outperforming DLR by 1.4% and the strongest single skill by 0.2%. This isolates the benefit of the final category-to-skill mapping while retaining a single video MLLM pass.

**Effect of routing set size.** To measure how many such samples are needed for routing table construction, we vary the routing-set size while keeping the discovered Top-5 toolbox and the inference configuration fixed. As shown in Table 7, AutoSkill is already effective with a small set: 300 samples give a +2.0% average gain over the baseline. Increasing the size further improves performance gradually, reaching +2.2% around 400–600 samples and +2.3% at 800–1,200 samples. With 1,500 samples the average reaches +2.4%, after which additional samples bring only marginal returns. We also randomly resample and rewrite the 1,500-example routing set five times and rebuild the table independently. The variation across seeds is negligible, indicating that the learned mapping is stable and not tied to a particular rewrite. We therefore use  $N=1,500$  in the main experiments as a trade-off between routing reliability and construction cost.

**Why voting is not an effective router.** Majority voting over the Top-5 skills reaches 58.6%, matching the strongest single skill and remaining below AutoSkill, while requiring 5 video MLLM passes per sample. The reason is that for many instances only one or two skills recover the decisive evidence, so the correct answer is a minority vote and is overwritten by the majority of mismatched skills. Skill ensembling there-

Table 6: Computational cost versus RL post-training. Data denote SFT+RL for the baselines and labelled+unlabelled samples for ours.  $^\dagger$  uses training time reported in the official repository.  $^\ddagger$  marks discovery evaluation only.

| Method                         | Update | Data      | Cost                        |
|--------------------------------|--------|-----------|-----------------------------|
| Video-R1 (Feng et al. 2026)    | Yes    | 165K+260K | 440 A100-h                  |
| LongVILA-R1 (Chen et al. 2026) | Yes    | 36K+170K  | 9.2–12.3K A100-h $^\dagger$ |
| VideoZoomer (Ding et al. 2025) | Yes    | 52K       | ~768 H100-h                 |
| AutoSkill (ours)               | No     | 3K+1.5K   | ~30 A100-h $^\ddagger$      |

Table 7: Routing-set size ablation on Qwen2.5-VL-7B (mean over 5 random rewrites from 1,500 target-style samples). LongVB denotes LongVideoBench.

| Samples  | MLVU        |             | LongVB      | VideoMME | LVBench     | Avg/ $\Delta$      |
|----------|-------------|-------------|-------------|----------|-------------|--------------------|
|          | dev         | test        |             |          |             |                    |
| baseline | 66.0        | 47.7        | 61.2        | 64.6     | 42.5        | 56.4 / –           |
| 300      | 68.1        | 50.8        | 61.7        | 65.2     | 46.2        | 58.4 / +2.0        |
| 400      | 68.5        | 51.2        | 61.7        | 65.1     | 46.3        | 58.6 / +2.2        |
| 500      | 68.6        | 51.1        | 61.9        | 65.2     | 46.4        | 58.6 / +2.2        |
| 600      | 68.6        | 51.1        | 61.8        | 65.2     | 46.6        | 58.6 / +2.2        |
| 800      | 68.7        | 51.2        | 61.8        | 65.2     | 46.5        | 58.7 / +2.3        |
| 1000     | 68.8        | 51.1        | 61.8        | 65.2     | 46.6        | 58.7 / +2.3        |
| 1200     | 68.9        | 51.2        | 61.8        | 65.1     | 46.7        | 58.7 / +2.3        |
| 1500     | <b>69.0</b> | <b>51.1</b> | <b>61.9</b> | 65.2     | <b>46.8</b> | <b>58.8 / +2.4</b> |

fore cannot exploit specialization when the useful skill is outnumbered. AutoSkill instead selects that specialized skill from the query category and answers with a single video MLLM pass.

## Efficiency Analysis

Table 6 contrasts AutoSkill with RL-based video post-training. Our discovery uses  $T=6$  cycles and  $M_c=5$  evaluations per cycle (about 30 runs). Each run takes roughly one A100 hour on the 300-sample development set, for about 30 A100-h in total, or about  $14.7\times$  less than the 440 A100-h reported by Video-R1 (Feng et al. 2026). This estimate excludes routing-table construction. Under the matched protocol, AutoSkill gains +2.4% without weight updates, and at inference it uses one video MLLM pass plus a text-only routing call.

## Conclusion

We presented AutoSkill, a framework that replaces one universal frame selector with an automatically discovered skill toolbox and taxonomy-guided category-to-skill routing. An LLM agent iteratively develops skills on a capability-focused development set of a frozen video MLLM, then adapts the toolbox to each target domain without using target answers or updating model weights. Experiments show that the discovered skills provide complementary gains and that the final router outperforms both the discovery-cycle router and



the strongest single skill. AutoSkill therefore turns agent-driven exploration into an efficient and deployable approach for long-video question answering.

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